Two neighboring crossing kernels and an asymmetric larger-noise comparison

Research note, 28 October 2026

The nonnegative kernels in the general crossing identity allow neighboring counts to be combined before estimating their probabilities. This eliminates all but one incoming count from the lower bound below. As an application we prove the central comparison through $\lambda = .36$ when exactly one conditional law has two-point noise. This is a restricted theorem about actual iid laws, not a replacement for the general larger-noise comparison.

1 A two-count reserve

Use the notation and hypotheses of the retained crossing theorem: U_0, U_1 are centered, independent conditional laws,

$$-A_0 \le U_0 \le a_0, \quad -a_1 \le U_1 \le A_1, \qquad a_0 + a_1 < 1, \quad A_1 + a_1 \le 1.$$

For mutually independent copies put

$$W_j = \sum_{l \le j} U_{1l} + \sum_{l > j} U_{0l}, \quad T_j = j + W_j, \quad L_j(t) = \mathbb{E}(t - T_j)_+.$$

Assume $na_0 \le t < n(1 - a_1)$ and write

$$g_i(t) = L_{i-1}(t) - L_i(t) - \mathbb{P}(T_i \le t) \ge 0.$$

Nonnegativity is the analytic conclusion of the crossing theorem, not an arithmetic assumption imposed on test cases.

Theorem 1 (Paired reserve)**.** *For any integer $1 \le k \le n - 1$, put $u = t - k$. Then*

$$\begin{aligned} g_k + g_{k+1} = u + \mathbb{P}(W_k > u) + \mathbb{E}(W_{k-1} - u - 1)_+ \\ - \mathbb{E}\big[(u - W_{k+1})\mathbf{1}_{\{W_{k+1} \le u-1\}}\big]. \end{aligned} \tag{1}$$

In particular, if Q bounds the last expectation and $\pi_j = \binom{n}{j} p^j (1-p)^{n-j}$, then

$$\sum_{i=1}^{n} \pi_i g_i \ge \min\{\pi_k, \pi_{k+1}\}\{u + \mathbb{P}(W_k > u) - Q\}. \tag{2}$$

The incoming event in these formulas is closed.

Proof. Add the two local identities to obtain $g_k + g_{k+1} = L_{k-1} - L_{k+1} - \mathbb{P}(T_k \le t) - \mathbb{P}(T_{k+1} \le t)$. Centering and put–call parity give $L_{k-1} = u + 1 + \mathbb{E}(W_{k-1} - u - 1)_+$. Also $L_{k+1} + \mathbb{P}(T_{k+1} \le t)$ equals the last expectation in (1), including an atom at $W_{k+1} = u - 1$, where its weight is one. This proves the identity. Drop its nonnegative outgoing stop-loss and then use $g_i \ge 0$ to prove (2). No independence of W_k and W_{k+1} is asserted or needed. □

Lemma 2 (One hinge bounds the weighted incoming event)**.** *Let $b = 1 - u$ and choose $y < b$ with $y \geq -1 - u$. For every real z,*

$$(u + z)\mathbf{1}_{\{z \geq b\}} \leq \frac{(z - y)_+^2}{(b - y)^2}.$$

Thus any justified quadratic-hinge bound for $Z = -W_{k+1}$ gives a bound on Q in Theorem 1 without an additional factor. Use the raw hinge quotient: clipping it at one as for an ordinary probability is not generally valid for this weighted expectation.

Proof. For $z \geq b$, the derivative of $(u+z)/(z-y)^2$ has the sign of $-(z+y+2u)$. This is nonpositive because $b + y + 2u = 1 + u + y \geq 0$. The ratio at b is $1/(b-y)^2$. Below b the desired left side is zero. □

There is also a reduction to one conditional sum. With $d = a_0 + a_1$ and $\zeta = 1 - d$, the auxiliary coupling gives $W_{k+1} \geq W_k - d$ pathwise. Consequently

$$g_k + g_{k+1} \geq u + \mathbb{E}\left[\mathbf{1}_{\{W_k > u\}} - (u - W_k + d)\mathbf{1}_{\{W_k \leq u - \zeta\}}\right]. \tag{3}$$

Indeed the incoming event implies $W_k \leq u - \zeta$, and its weight is at most $u - W_k + d$. The affine left tail in (3) must be kept; a polynomial for the indicator alone is not a valid minorant of it.

2 An elementary lower bound for the central binomial masses

Lemma 3. *If $n \geq 42$, $p \in [2/5, 53/100]$ and $k = np$ is an integer, then*

$$\sqrt{n}\min\{\pi_k, \pi_{k+1}\} > \frac{1}{2}, \qquad k \geq 17, \quad n - k \geq 20.$$

Proof. For any integer-valued variable of variance V and largest atom M, add an independent uniform variable on $[-1/2, 1/2]$. The resulting density is at most M and has variance $V + 1/12$. A density bounded by M has variance at least $1/(12M^2)$: about its mean its tail probability is at least $(1 - 2Mr)_+$, and integrate $2r(1 - 2Mr)_+$ over $r \geq 0$. Thus $M \geq 1/\sqrt{12V + 1}$.

For the binomial law with integer mean, π_k is its largest mass and $\pi_{k+1}/\pi_k = k/(k+1) \geq 17/18$. Since $p(1-p) \leq 1/4$,

$$\sqrt{n}\min\{\pi_k, \pi_{k+1}\} \geq \frac{17}{18\sqrt{3 + 1/42}} > \frac{1}{2}.$$

The last strict inequality is the rational comparison $289/81 > 127/42$. The integer bounds follow from $np \geq 16.8$ and $n(1-p) \geq 19.74$. □

3 The complete comparison for two asymmetric families

Write $v = p(1-p)$, $s = (1-p)\mathbb{E}U_0^2 + p\mathbb{E}U_1^2$, $\lambda = ns$, and $Y = B + U$. Put

$$\rho = \frac{\mathbb{E}|Y - p|^3}{(v + s)^{3/2}}, \qquad \rho_B = \frac{1 - 2v}{\sqrt{v}}.$$

At the central integer anchor $t = k = np$ the normal terms agree exactly. The direct comparison is therefore

$$\Delta_C = \sqrt{n}\{\mathbb{P}(Y_1 + \cdots + Y_n \leq k) - F_B(k)\} + C(\rho_B - \rho).$$

The signed-curvature lemma from the retained v13 note, whose hypothesis $t \geq np - (1-p)$ holds here, gives

$$\Delta_C \leq -\sqrt{n}\sum_i \pi_i g_i + \frac{\lambda}{4v\sqrt{n}} + C(\rho_B - \rho). \tag{4}$$

Theorem 4 (Central asymmetric comparison through $\lambda = .36$)**.** *Let $n \geq 42$, $p \in [2/5, 53/100]$, $k = np \in \mathbb{Z}$, $A = 3/5$, and $0 < \lambda \leq A^2 = 9/25$. Consider either of the following centered conditional laws, with the probabilities uniquely determined by centering:*

1. *$U_0 \in \{-A, \varepsilon\}$, $U_1 = 0$, where $\varepsilon = \lambda/((n-k)A)$;*
2. *$U_0 = 0$, $U_1 \in \{-\varepsilon, A\}$, where $\varepsilon = \lambda/(kA)$.*

For every $C \geq 0$ the central comparison is strict. More precisely,

$$\Delta_C < -\frac{1}{240} \quad \text{in case 1,} \qquad \Delta_C < -\frac{25}{48}\lambda \quad \text{in case 2.}$$

Both families satisfy the right-strip bounds $A = .6$, $d \leq .072$ and $H = 0$. The theorem includes $\lambda = .36$ and exact atom collisions.

Proof. We first check the cubic term rather than dropping it without a sign. For either family the inward support is at most $A/17 < .072$. More generally, within the right strip with $d \leq .072$, the signs of $Y - p$ on its two conditional supports are fixed. Expanding the two cubes, using $\mathbb{E}U_0^3 \leq a_0 \mathbb{E}U_0^2$ and $\mathbb{E}U_1^3 \geq -a_1 \mathbb{E}U_1^2$, gives

$$\begin{aligned}(v+s)^{3/2}(\rho - \rho_B) &\geq (3\min(p, 1-p) - d)s \\ &\quad - (1-2v)\left(\frac{3}{2}s + \frac{3s^2}{8v}\right) \\ &\geq s\left(\frac{87}{250} - \frac{13}{16}s\right) > 0.\end{aligned}$$

Here $(1+x)^{3/2} \leq 1 + 3x/2 + 3x^2/8$ for $x \geq 0$, $v \geq 6/25$ and $s \leq (9/25)/42$. Thus $C(\rho_B - \rho) \leq 0$.

In case 1 put $m = n - k \geq 20$ and $\mu = \lambda/A^2 \in (0, 1]$, so $\varepsilon = A\mu/m$. If J counts the rare negative jumps in the own count, then

$$W_k = m\varepsilon - (A + \varepsilon)J, \quad J \sim \mathrm{Bin}\left(m, \frac{\mu}{m+\mu}\right).$$

Consequently $W_k > 0$ exactly when $J = 0$, and

$$\mathbb{P}(W_k > 0) = (1 + \mu/m)^{-m} > e^{-\mu} \geq e^{-1} > 1/3. \tag{5}$$

The last elementary inequality follows from $e < 3$. For the incoming count put $c = A + \varepsilon$, $b = (m-1)\varepsilon$; then $Z = -W_{k+1} = cJ - b$ with $J \sim \mathrm{Bin}(m-1, \mu/(m+\mu))$ and $Q = \mathbb{E}[Z\mathbf{1}_{\{Z \geq 1\}}]$.

If $0 < \mu \leq 2/5$, the incoming event requires $J \geq 2$, because $c - b = A - (m-2)\varepsilon < 1$. Hence

$$Q \leq c\mathbb{E}J(J-1) \leq A\mu^2, \qquad \mathbb{P}(W_k > 0) \geq 1 - \mu.$$

The latter follows from the union bound on the own rare jumps. Therefore $\mathbb{P}(W_k > 0) - Q \geq 1 - 2/5 - (3/5)(2/5)^2 = 63/125 > 2/15$.

If $2/5 \leq \mu \leq 1$, then

$$2c - b = A\{2 - \mu(1 - 3/m)\} \leq 249/250 < 1,$$

so the incoming event requires $J \geq 3$. For $j \geq 3$, $(cj - b)/(j(j-1))$ decreases with j, since its consecutive difference has the positive numerator $c(j+1) - 2b$. Thus

$$\begin{aligned}Q &\leq \frac{3c - b}{6}\mathbb{E}J(J-1) \\ &= \frac{A}{6}\frac{\mu^2\{3 - (1 - 4/m)\mu\}(1 - 1/m)(1 - 2/m)}{(1 + \mu/m)^2} \leq \frac{A}{3} = \frac{1}{5}.\end{aligned} \tag{6}$$

For completeness, the μ-dependent fraction increases on $[0,1]$: its derivative has the positive numerator $\mu\{6-3(1-4/m)\mu-(1-4/m)\mu^2/m\}$. At $\mu=1$, the final inequality reduces, with $h=1/m$, to $(1+2h)(1-h)(1-2h)\le(1+h)^2$; their difference is $3h+5h^2-4h^3\ge 0$. Combining (5) and (6) yields the paired reserve $>2/15$. Theorem 1 and Lemma 3 give $\sqrt{n}\sum_i \pi_i g_i > 1/15$. Since $v\ge 6/25$ and $\sqrt{n}>6$,

$$\frac{\lambda}{4v\sqrt{n}} < \frac{9/25}{4(6/25)6} = \frac{1}{16}.$$

Equation (4) proves the first conclusion.

In case 2 put $\mu=\lambda/A^2$, so $\varepsilon = A\mu/k$. Now $W_k=(A+\varepsilon)J-k\varepsilon$ and $W_k>0$ exactly when $J\ge 1$. Bernoulli's inequality gives

$$\mathbb{P}(W_k>0) = 1-(1+\mu/k)^{-k} \ge \frac{\mu}{1+\mu} \ge \frac{\mu}{2}.$$

Moreover $-W_{k+1}\le (k+1)\varepsilon \le 54/85<1$, so $Q=0$. Therefore the normalized paired loss is greater than $\mu/4$. Again using (4),

$$\Delta_C < -\frac{\lambda}{4A^2}+\frac{\lambda}{4(6/25)6} = -\frac{25}{48}\lambda.$$

All threshold and support assumptions hold: the inward bound is $\varepsilon\le A/17$, and $t=np$ lies between na_0 and $n(1-a_1)$. □

4 A bounded selected-noise bias

The following observation is useful independently of the two-point families. Let K be the unique crossing index from the retained crossing theorem, and let $\zeta = 1-a_0-a_1$. If also

$$A_0+a_0<\zeta, \qquad A_1+a_1<\zeta,$$

then

$$0\le \mathbb{E}W_{K+1} \le \mathbb{E}(U_0)_+ \le a_0, \qquad 0\le -\mathbb{E}W_K \le \mathbb{E}(U_1)_- \le a_1. \tag{7}$$

To see this, the leave-pair residuals $R_i = t-(i-1)-\sum_{l<i}U_{1l}-\sum_{l>i}U_{0l}$ decrease by at least ζ on every sample. The truncated centered mean $M_0(x)=\mathbb{E}[U_0\mathbf{1}_{\{U_0\le x\}}]$ vanishes outside $[-A_0,a_0)$ and satisfies $0\le -M_0(x)\le \mathbb{E}(U_0)_+$. At most one residual lies in that interval. The exact bias identity $\mathbb{E}W_{K+1} = -\sum_i \mathbb{E}M_0(R_i)$ proves the first assertion. For the other sign, write $W_K = \sum_i[U_{1i}\mathbf{1}_{\{T_i\le t\}}+U_{0i}\mathbf{1}_{\{T_i>t\}}]$. The second term has mean zero because U_{0i} is independent of T_i. Thus $\mathbb{E}W_K=\sum_i \mathbb{E}M_1(R_i-1)$. Its active interval is $[1-a_1,1+A_1)$, proving the second assertion. These half-open intervals include lower-endpoint atoms; the upper endpoint contributes zero by centering. The acquired right strip has both width bounds, since each width is at most $.672<.928\le\zeta$.

5 Scope, dependencies and tests

The paired identity uses the general crossing theorem in `crossing.tex` in the v13 research notes (28 October 2026). The normalized family theorem additionally uses the signed-curvature lemma in the adjacent `counts-tools.tex`; its proof there handles all counts and finite sample sizes. The present proof of the family theorem uses no external Poisson comparison and no numerical grid.

The companion exact checks test the identities, the same-sum inequality, weighted quadratic hinges, the selected-bias caps and actual family comparisons, including rare jumps and collision endpoints. They preserve negative controls for a missing closed collision, an inadmissible hinge, and unjustified clipping of a weighted expectation at one. The acceptance gates are finite algebraic checks of the inequalities displayed above, not samples presented as uniform coverage.

No conclusion here removes the general count certificates. The proved family requires $H=0$, np integral and one noiseless conditional law; general conditional laws and nonzero phases still need a uniform lower bound on the paired weighted reserve, or another way to absorb curvature.